

GABRIEL CARATIHAN

SOFTWARE ENGINEER & SYSTEMS ARCHITECT

Email: gabrielcaratihan2003@gmail.com | GitHub: github.com/GabbyDev0402 | Location: Imus, Cavite, PH

EXECUTIVE SUMMARY

Software Engineer passionate about architecting full-stack, highly scalable SaaS applications that eliminate administrative friction and bridge the gap between operational bottlenecks and digital solutions. Former ESL Educator turned Systems Architect with dual perspective in institutional workflows and cloud software design.

CORE TECHNICAL SKILLS

Languages & Frontend

- TypeScript / JavaScript
- React / Next.js (App Router)
- Tailwind CSS v4 / HTML5
- Framer Motion

Backend & Cloud

- Supabase / PostgreSQL
- Node.js / REST APIs
- Firebase / Cloud Firestore
- Netlify / Vercel

Engineering Workflows

- Antigravity IDE (Gemini 3.5)
- Agentic Prompt Engineering
- RLS & DB Triggers
- Git & CI/CD Workflows

&i AGENTIC ENGINEERING WORKFLOW

Architected and engineered applications using advanced agentic workflows via the Antigravity IDE. Directed AI models (Gemini 3.5 Flash / 3.1 Pro) to rapidly prototype UI/UX, establish PostgreSQL & NoSQL data schemas, enforce Row Level Security (RLS) policies, and generate automated Playwright E2E testing suites.

EDUCATION & PROFESSIONAL BACKGROUND

Computer Programming Practitioner

Imus Computer College — Systems & Software Development

ESL Educator

Washington School — Educational Technology & Workflows

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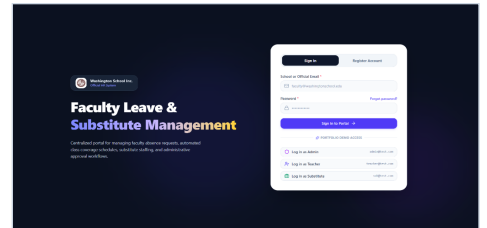
FEATURED ENGINEERING PROJECTS

1. Faculty Leave & Substitute Portal (NEW & FEATURED)

Enterprise Supabase HR Workflow Engine & Substitute Coverage Portal

Live App: <https://faculty-leave-management-portal.netlify.app/> | GitHub: <https://github.com/GabbyDev0402/faculty-leave-hr-portal>

- Engineered PostgreSQL Row Level Security (RLS) policies and database triggers for automated leave balance deduction.
- Built relational one-to-many schema for claiming class blocks and 60-second cryptographic signed URLs for lesson plans.
- Integrated real-time Postgres WebSocket synchronization and dynamic Next.js layout overrides for /login.

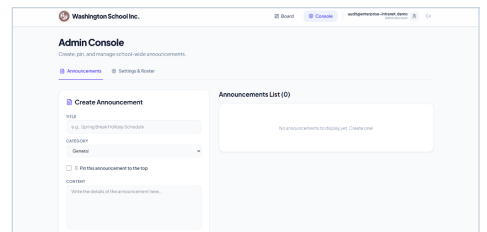


2. Enterprise Communications Intranet

B2B Announcement & Administrative Compliance Portal

Live App: <https://washington-school-portal.netlify.app/admin> | GitHub: <https://github.com/GabbyDev0402/washington-school-portal>

- Implemented Role-Based Access Control (RBAC) with Cloud Firestore Security Rules.
- Engineered digital 'Read Receipt' compliance tracking and automated client-side CSV audit log reporting.
- Streamlined institutional announcement workflows with secure multi-role administrative oversight.

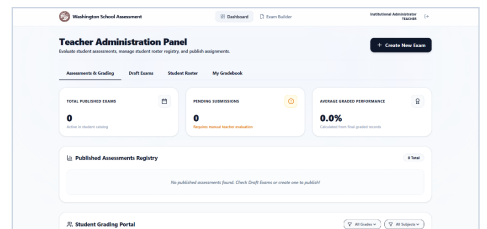


3. Washington Assessment Portal

Multi-Tenant LMS with Automated Exam Engine & Analytics

Live App: <https://wcs-exam-portal.netlify.app/> | GitHub: <https://github.com/GabbyDev0402/washington-school-portal>

- Architected dynamic exam creation engine utilizing polymorphic React form components.
- Engineered real-time pivot-table data aggregation for Master Gradebooks and teacher performance metrics.
- Enforced strict multi-tenant data isolation and time-gated client routing for exam security.



4. AttendancePro Tracker

Proactive Attendance Management & Truancy Early-Warning Engine

Live App: <https://wcsattendancetracker.netlify.app/> | GitHub: <https://github.com/GabbyDev0402/wcs-attendancetracker>

- Optimized NoSQL database reads using session-grouped data modeling for rapid attendance logging.
- Designed proactive early-warning algorithms to detect at-risk student truancy patterns.
- Engineered automated instruction-loss calculations to assist school administrators with intervention.

